

The Effect of Contamination on Contamination Control Regulation

US Coal Mining and Drinking Water Utilities

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Self-reported compliance

- Self-reported compliance: entity monitors and reports themselves to regulators.
- Incentives to shirk reporting:
 - ▶ Concealing non-compliance
 - ▶ Saving self-reporting costs
- Perversely, incentives to under-report increase with contamination for environmental contamination regulation (e.g. US drinking water).
- **Why care about self-reporting and contamination?**
 - ▶ Reporting is most valuable to regulators when contamination is rising
 - ▶ Different fix if concealing non-compliance vs self-reporting costs
 - ▶ 85% of US household drinking water is regulated this way
 - ▶ Self-reporting used across industries (HIPAA violation disclosure, SOX, energy firms)

Research question

1. What happens to firm self-reporting when contamination rises?
2. Are changes in self-reporting due to concealment or self-reporting costs?

Context: US drinking water utilities exposed to water contamination from upstream coal mining (1985–2005)

0. Does upstream coal mining raise contamination at drinking water utilities?

How I answer these questions

- Annual panel of drinking water utilities on average 22 km downstream of active coal mines between 1985 and 2005.
- Separation between contamination source and regulated firm helps identify the effect of contamination.
- Coal mining → inorganic chemicals (“IOCs”) into surface and ground water.
- Contaminant concentrations for several IOC’s 1998–2005 link coal production to contamination.
- Instrument upstream coal production.
- Utility self-reporting, regulator visits, and enforcement received.

Utility nitrate self-reporting and under-reporting incentives

Utility manager/staff cognitive burden (Oleckno, 1982).

Self-reporting requirement for nitrates (one of 83 contaminants):

- **Baseline:** once a year, send one sample from every intake to a lab
- **Elevated:** nitrate ≥ 5 mg/L, half the Maximum Contaminant Level (“MCL”), test quarterly

Miss a test, receive a monitoring and reporting (“MR”) violation and:

- Avoid an MCL (need self-report for MCL)
- MR punished less severely than MCL
- Regulator visit
- Or enforcement: Informal (letters, notices, compliance plans); Formal (consent decrees, fines, prosecution)

Is coal production a source of contamination at utilities?

$$Concen_{ct} = \beta \cdot \sum_{\tau=1985}^t \text{UpstreamCoalProd}_{c\tau} + \rho_c + \text{Watershed}_c \cdot \tau_t + \epsilon_{ct}$$

- $Concen_{ct}$: mean annual concentration at utility c , year t (SYR2, 1998–2005).
- β : effect of 1 million additional cumulative short tons mined upstream since 1985.
- ρ_c : utility fixed effects. $\text{Watershed}_c \cdot \tau_t$: Watershed-specific year effects, absorbing watershed-wide industrial trends.
- SEs clustered at the utility level.
- Continuous treatment, **no control**: identification comes from relative exposure.
- Assumption: utilities with large upstream production changes would have trended like those with small ones. (balance test on utility characteristics $\rightarrow$)

Mining raises contaminant levels

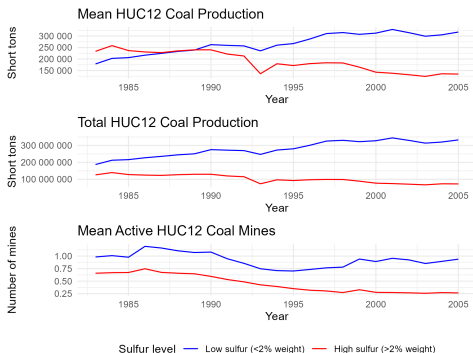
	Arsenic (mg/L) (1)	Nitrate (mg/L) (2)	Barium (mg/L) (3)	Selenium (mg/L) (4)
Cumul. upstream coal prod. (M ST)	0.00001 (0.00001)	0.00883*** (0.00257)	0.00136 (0.00112)	0.00004** (0.00002)
Sample mean (million short tons)	6.9	8.6	6.7	6.7
MCL (mg/L)	0.050	10.000	2.000	0.050
Observations	366	472	353	350

Notes: All specifications include utility and watershed $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

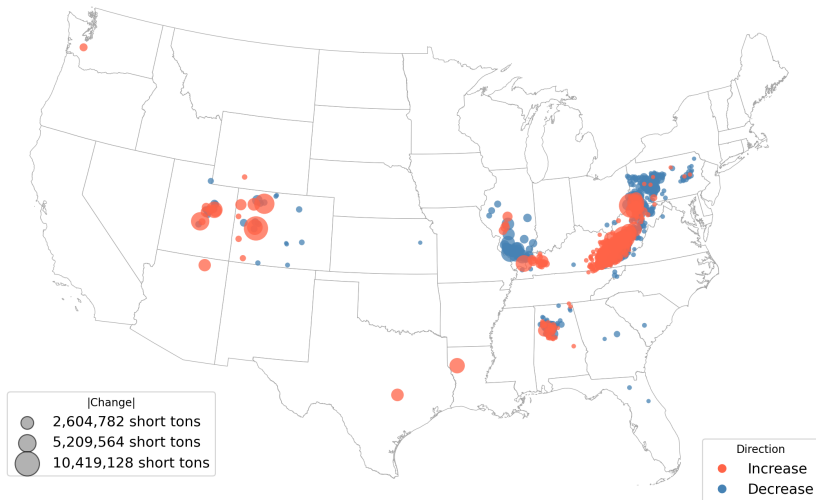
Every reading in the sample stays under its MCL. In the sample, only 49 IOC MCL violations.

Does mining increase under-reporting? Instrumenting coal mining

- OLS: mining and utility characteristics not random.
- Adapt Douglas and Wiggins (2015) Acid Rain Program (“ARP”) instrument for coal production.
- ARP set coal fired power plant SO_2 limits.
- Low sulfur coal was the main compliance response (Lattanzio, 2022).
- Use coal sulfur % and the 1995 SO_2 power plant compliance deadline.



Where coal production moved, 1985–2005



2SLS

First stage:

$$\text{NumMines}_{ct} = \alpha_0 + \alpha_1 X_{ct} + \alpha_2 \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \\ + \text{State}_s \cdot Y_t + C_c + \nu_{ct}$$

Second stage:

$$\text{Outcome}_{vct} = \beta_0 + \beta_1 X_{ct} + \gamma \widehat{\text{NumMines}}_{ct} + \text{State}_s \cdot Y_t + C_c + \epsilon_{ct}$$

- $\text{Post95}_t = \mathbf{1}\{t \geq 1995\}$. Sulfur is time-invariant, so C_c absorbs its level.
- X is time varying utility characteristics.
- Outcome: 1 if utility c had a violation, visit, or enforcement of type v at any point in year t .
- Coefficients are in percentage points.
- SEs clustered at the utility level.

Instrument validity

- **Relevance.** First stage F-stat 49.2.
- **Independence.** The ARP is a national clean-air law set by national energy and emissions politics, not downstream utilities' water quality.
- **Exclusion.** Effect of instrumenting mining downstream of the utility on utility behavior is not significant. (MR violations →), (visits →), (enforcement →).
Utility characteristics are not predicted by the instrument. (full balance table →)
- **Monotonicity.** A defier is a high-sulfur watershed that expanded *because* its coal was high sulfur. Scrubber adoption costs more likely lead to lost rather than gained market share; low-cost high-sulfur regions that expand anyway are never-takers.

Upstream mining reduces utilities' self-reporting

	Monitoring and reporting (MR) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal mines (sum)	0.68** (0.33)	0.55 (0.34)	0.62** (0.31)	0.42 (0.33)	0.49 (0.32)	0.19 (0.34)
2SLS						
Upstream coal mines (sum)	5.12*** (1.70)	3.94** (1.75)	4.85*** (1.57)	3.78** (1.55)	4.66*** (1.62)	3.06* (1.66)
RF						
Post-1995 × Upstream sulfur %	-2.12*** (0.66)	-1.95** (0.83)	-2.01*** (0.60)	-1.87** (0.73)	-1.93*** (0.63)	-1.51* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓

Notes: The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Base rates of nitrates, arsenic, and IOC are 5.7%, 3.7%, and 4.3%.

The regulator increases inspections

	Sanitary (1)	Technical assistance (2)	Enforcement (3)	Sample collection (4)	Inspection (5)
2SLS					
Upstream coal mines (sum)	0.78 (1.20)	0.67 (0.95)	1.07 (0.81)	0.48 (0.50)	2.13** (1.07)

Notes: All specifications include utility and state $\times$ year fixed effects. The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Baseline inspection rate is 3.4%.

Despite inspection increase, enforcement does not increase

	Informal		Enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
2SLS						
Upstream coal mines (sum)	-3.06 (2.15)	-0.71 (2.10)	-3.37*** (0.95)	-1.08 (0.86)	2.37 (2.11)	-0.42 (2.25)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓

Notes: The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Baseline enforcement rates: informal 18.3%, formal 2.9%.

Inspections would reveal concealment and increase penalties - which we do not see.

Conclusion

- Contamination reduces self-reporting.
- Contaminant data and increases in inspections without increasing enforcement suggests that under-reporting is not concealing MCLs.
- Evidence of cost driven under-reporting.
- Regulators' inspection visits substitute for utility self-reporting.
- MR violations are not strongly punished.
- **Policy:** raising self-reporting penalties or reducing self-reporting costs could reduce under-reporting.

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Appendix

Appendix: instrument balance on intake facilities

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		0.9942*** (0.3834)
Post-1995 $\times$ Upstream sulfur	0.0509 (0.0386)	0.0563 (0.0376)
Utility fixed effects	✓	
Year fixed effects		✓
State $\times$ year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

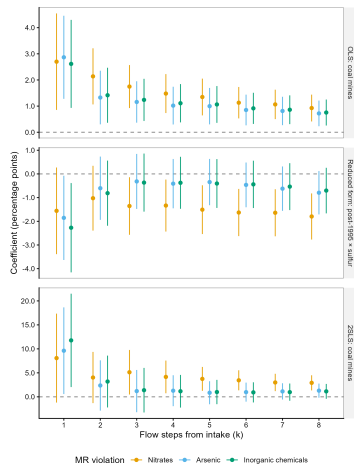
Appendix: balance test on utility characteristics

	Cumul. upstream coal prod. (10M ST)		Any upstream coal mining	
	(1)	(2)	(3)	(4)
Constant	0.2504 (0.2532)		-0.1596 (0.1198)	
Log backcast pop. 1997	0.1091 (0.0750)	0.1512 (0.0938)	0.0159 (0.0126)	0.0208 (0.0144)
Log pop. served (SYR2)	-0.0934 (0.0666)	-0.1113 (0.0956)	0.0060 (0.0259)	-0.0301 (0.0334)
Number of intake facilities	0.0089 (0.0569)	-0.0324 (0.0744)	0.0680* (0.0359)	0.0721* (0.0406)
Surface water source	0.1225 (0.1643)	-0.3902 (0.3296)	0.1210 (0.1307)	0.0521 (0.1601)
Public ownership	-0.1554 (0.2551)	-0.0599 (0.2240)	0.2084** (0.0911)	0.1048 (0.1098)
Wholesaler	-0.3780* (0.2041)	-0.7465 (0.9141)	-0.2164 (0.1340)	-0.2212 (0.1683)
Source water protection	-0.4549 (0.2749)	-0.4385 (0.4305)	-0.2807* (0.1521)	-0.1460 (0.1930)
Joint <i>F</i> -test (all covariates)	2.33	0.96	8.96	2.46
<i>p</i> -value	0.029	0.461	0.000	0.023
Observations	122	120	122	120
State fixed effects		✓		✓

Notes: Heteroskedasticity-robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

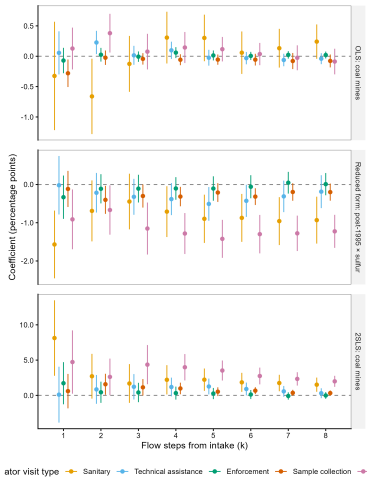
Observable characteristics jointly predict continuous upstream production only without state fixed effects ($p=0.029$) and lose significance once state fixed effects are added ($p=0.461$), as in the main specification. The joint test on the binary presence of upstream mining stays significant with or without state fixed effects.

Appendix: exclusion placebo, MR violations across k



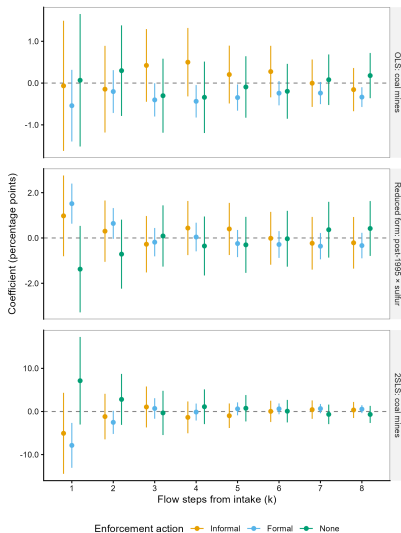
For utilities upstream of a mine — where mining cannot reach the intake hydrologically — OLS estimates on MR violations are positive and significant at every k . In the reduced form and 2SLS, nitrate estimates are significant for $k \geq 3$, while arsenic and inorganic chemical estimates are significant only at $k = 1$.

Appendix: exclusion placebo, visits across k



For utilities upstream of a mine, inspection visits rise significantly at nearly every k , sanitary visits at $k = 1$ and $k \geq 4$, and sample collection visits at several radii; technical assistance and enforcement visits are mostly insignificant.

Appendix: exclusion placebo, enforcement across k



Enforcement actions show no consistent pattern for utilities upstream of a mine at any choice of k , supporting the exclusion restriction.

References I



Douglas, Stratford and Seth Wiggins (2015). *Effects of Acid Rain Regulations on Production of Eastern Coals of Varying Sulfur Content*. College of Business and Economics Working Paper. West Virginia University.



Lattanzio, Richard K. (2022). *Clean Air Act: A Summary of the Act and Its Major Requirements*. Tech. rep. Washington, DC: Congressional Research Service.



Oleckno, William A. (1982). "The National Interim Primary Drinking Water Standards: Part II, Impact on Community Water Systems: The Indiana Experience". In: *Journal of Environmental Health* 44.5, pp. 240–244.